UNTRM AI WORKSHOP · SESSION 1

From Data to Agents: A Hands-On Introduction to AI

Edgar Lobaton, Ph.D. · Director of Applied AI, College of Engineering
Professor of Electrical and Computer Engineering, NC State University

Chachapoyas · Monday, November 9, 2026

NC STATE UNIVERSITY

ABOUT ME · FROM PERU

Edgar Lobaton, Ph.D.



UNDERGRADUATE

B.S. Mathematics
B.S. Electrical Engineering (EE), Seattle U.

→

DOCTORATE

Ph.D. in EE and Computer Sciences, UC Berkeley

→

POSTDOC

Department of Computer Science, UNC Chapel Hill

→

SINCE 2011

ECE faculty, NC State University

TODAY

Professor, Electrical and Computer Engineering · Director of Applied AI, College of Engineering · NAIRR AI Education Fellow, Computing Research Association

ACTIVE ROBOTIC SENSING (AROS) LAB

AI for living systems and health

Biological

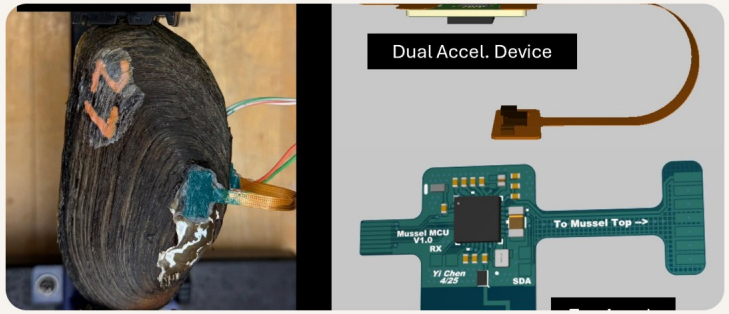
- Nematode, foram imaging
- Plant 3D phenotyping
- Mussel river sentinels

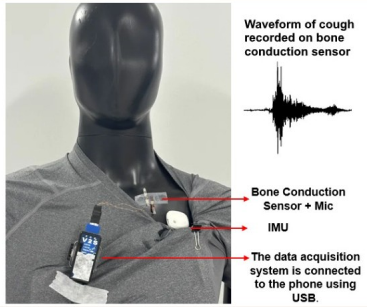
Biomedical

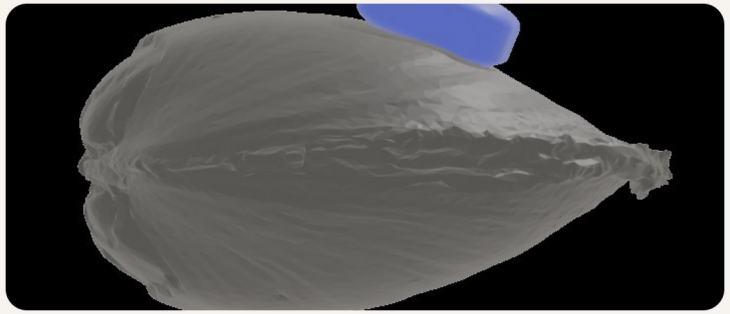
- Cough, asthma, apnea
- Dementia monitoring
- Robotic prosthetics

AI themes

- Robustness, uncertainty
- Embedded AI, robotics
- Geometry and physics
- Digital twins







THE WORKSHOP

Four sessions, two days

MONDAY · SESSION 1 · TODAY

From Data to Agents

What AI is, and how to specify, co-design and evaluate with it

MONDAY · SESSION 2

Inside the Black Box

How deep learning works: CNNs, RNNs and autoencoders

TUESDAY · SESSION 3

Coding with a Copilot

Adapt pre-trained models to your research with AI code assistants

TUESDAY · SESSION 4

Seeing with AI

A full image workflow, from annotation to deployment

Monday: no coding, in the browser · **Tuesday:** Python, on the university HPC and lab workstations

THIS SESSION

What you'll leave with

A mental model
How AI learns from data, and how context, tools and agents shape its answers

A way of working
Specify, co-design and evaluate, choosing your mode on purpose

Your own assistants
Two NotebookLM personas and a notebook on your own papers

An analysis you checked
A classifier planned with AI in Colab, and what it decided on its own

Everything runs in the browser with a free Google account. No coding needed for this session.

SHOW OF HANDS

Who's in the room?

Your area

- Agricultural data
- Computer vision and microscopy
- Genomics and biotechnology
- Other

Your role

- Faculty
- Researcher
- Graduate student
- Undergraduate
- Staff

SHOW OF HANDS

How do you use AI today?

How often?
Never · Occasionally · Weekly · Daily

Which tools?
ChatGPT · Gemini · Copilot · Claude · Other

For what?
Writing · Literature · Code · Data analysis

Do you program?
Python · R · Not yet

Has an AI ever given you a wrong answer with full confidence?

Today's plan

PART 1 · 20 MIN

What AI is

Show of hands

PART 2 · 45 MIN

Context, tools and agents

NotebookLM personas

PART 3 · 15 MIN

Your own documents

NotebookLM

PART 4 · 15 MIN

Working with AI

Modes, skills, learning

PART 5 · 20 MIN

Plan, co-design, evaluate

NotebookLM + Colab

WRAP-UP · 5 MIN

Take home

Checklist and next steps

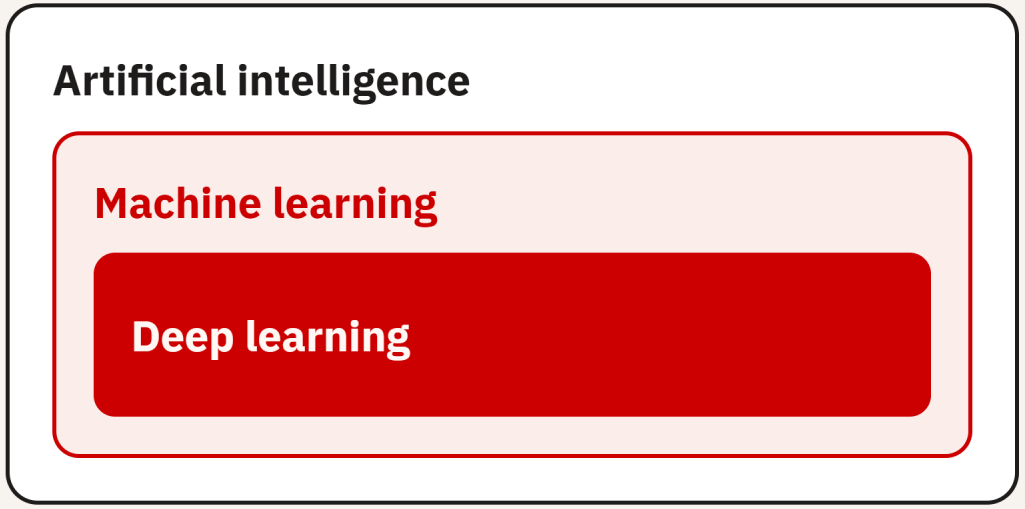
PART 1

What AI is

Three types of AI, three ways machines learn, and how text and images become numbers

NC STATE UNIVERSITY

AI, machine learning and deep learning



AI: computers doing tasks that usually need human intelligence.

ML: learns patterns from data instead of rules written by hand.

Deep learning: neural networks that discover the important features themselves.

Three types of AI

PREDICTIVE

What will happen?

A forecast, score or class, learned from past data

e.g., a yield forecast

GENERATIVE

What can be created?

New text, images, code or sequences, sampled from what it learned

e.g., drafting a report

AGENTIC

What should be done?

Multi-step tasks, using reasoning and tools over several rounds

e.g., run an analysis, check it, revise

SHOW OF HANDS

Predictive, generative or agentic?

1. Count spores in microscope images
2. Write a technical bulletin for farmers about a crop disease
3. Estimate coffee yield per plot from rainfall, altitude and soil data
4. A system searches genome databases on its own, runs comparisons and writes a report
5. AlphaFold predicts a protein's 3D structure from its sequence
6. A drone detects disease in a field and schedules spraying

And for question 6: who should approve the spraying?

Three ways machines learn

SUPERVISED

What is the output for this input?

Learns from labeled examples

e.g., crack detection from labeled images

UNSUPERVISED

What patterns exist in this data?

Finds groups or structure without labels

e.g., clustering pump-failure signatures

REINFORCEMENT

Which action earns the best reward?

Learns by trial and error

e.g., a robot learning to walk

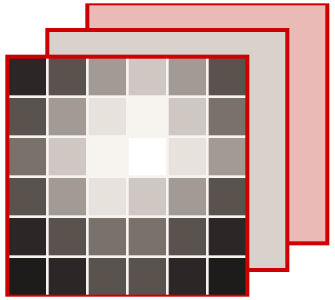
SHOW OF HANDS

Supervised, unsupervised or reinforcement?

1. An irrigation controller learns by trial and error
2. Find samples with unusual gene expression
3. Photos of leaves labeled healthy or diseased
4. A robot learns to stay balanced in simulation
5. Group soil samples by microbial profile, with no labels

To a computer, everything is numbers

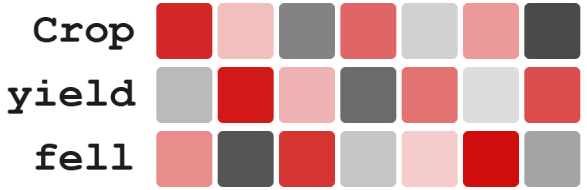
Image



A grid of pixel values for each color channel

`(224, 224, 3)`

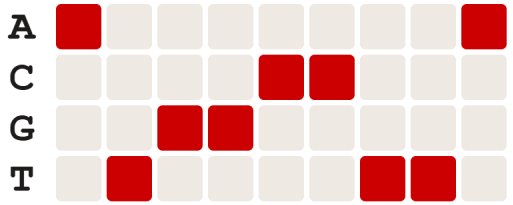
Text



Each token becomes a vector of learned numbers

`(3 tokens, 768)`

DNA

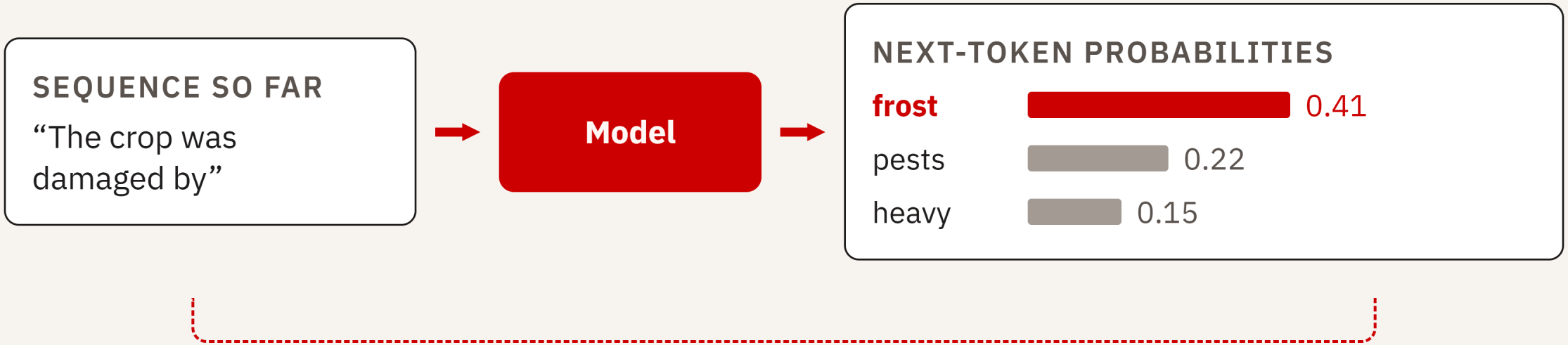


ATGGCCTTA: one column per base, one filled cell

`(9 bases, 4)`

Once the data are numbers with a known shape, the same kinds of models work across fields.

Generating text: predict the next token, again and again



Append the chosen token and repeat · images work similarly: remove a little noise, repeat (diffusion)

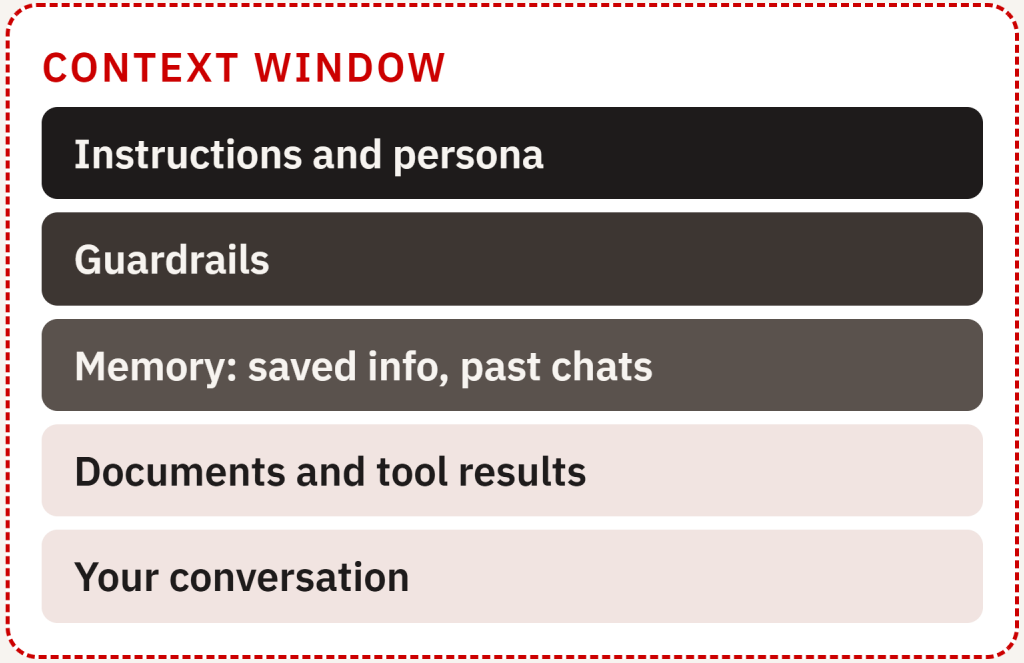
PART 2

Context, tools and agents

What the model sees, what it can do, and how several agents work together

NC STATE UNIVERSITY

The model only knows what is in its context



Everything the model “knows” right now lives in **one limited space**.

You control most of it.

What is missing, it fills in with plausible guesses.

INSTRUCTIONS, PERSONAS AND GUARDRAILS

Personas shape behavior, not knowledge

Same question: “My model works on lab images but fails on field photos. Why?”

Patient tutor

“Let’s think it through. What changed in the field photos: light, camera or background?”

Terse reviewer

“Distribution shift. Add field images to training or augment for lighting.”

PERSONA	“You are a patient tutor. Never give the answer directly; ask one guiding question at a time.”
LANGUAGE	“Always answer in Spanish, but keep technical terms such as ‘dataset’ in English.”
GUARDRAIL	“Never ask for or repeat personal data about farmers or patients.”

MEMORY

What the model carries from chat to chat

Saved info

Facts you asked it to remember about you and your work

Past chats

Earlier conversations it can look back at

The current chat

Everything said so far, until the window fills up

Memory has a size limit. The context window holds a fixed number of tokens. When it fills, older parts are dropped or summarized, and saved info keeps only short notes, not everything you said.

Memory is convenient, but it can also carry mistakes forward. You can review, edit or turn it off in settings.

WHEN MEMORY AND CONTEXT FAIL

Hallucinations: confident, not necessarily correct

Why it happens

- It predicts plausible text
- No built-in “I don’t know”
- Wrong or outdated memory carries forward
- Long chats lose details

Highest risk

- Citations and references
- Numbers and units
- Gene, protein and species names
- Doses and protocols

What helps

- A fresh chat for each new task
- Ask for sources, then open them
- Review or turn off memory
- Ground it in your documents (Part 3)

ACTIVITY A · 15 MIN

notebooklm.google.com

Same question, two personas

Set up each notebook: New notebook → add the beans **README** as a source → Configure chat → Custom → paste the persona.

Notebook 1 · Analysis Planner

"You are an analysis planner. Before proposing anything, ask me three clarifying questions. Then give a step-by-step plan and list your assumptions."

Notebook 2 · Critical Reviewer

"You are a critical reviewer. Do not solve the task. Point out the weakest assumption and the biggest risk in my request, and how to test each."

ASK BOTH

"I want a model these leaf photos into healthy, angular leaf spot and bean rust. How should I approach it?"

Compare: which answer helps you more, and which could you check?

Files: beans README.md and beans128.npz (shared folder)

TOOLS

Tools let a model do more than generate text

The model alone

- Only predicts text
- Knows nothing after its training
- Does arithmetic by guessing
- Can’t see your data

Search
Current information

Code
Exact calculations

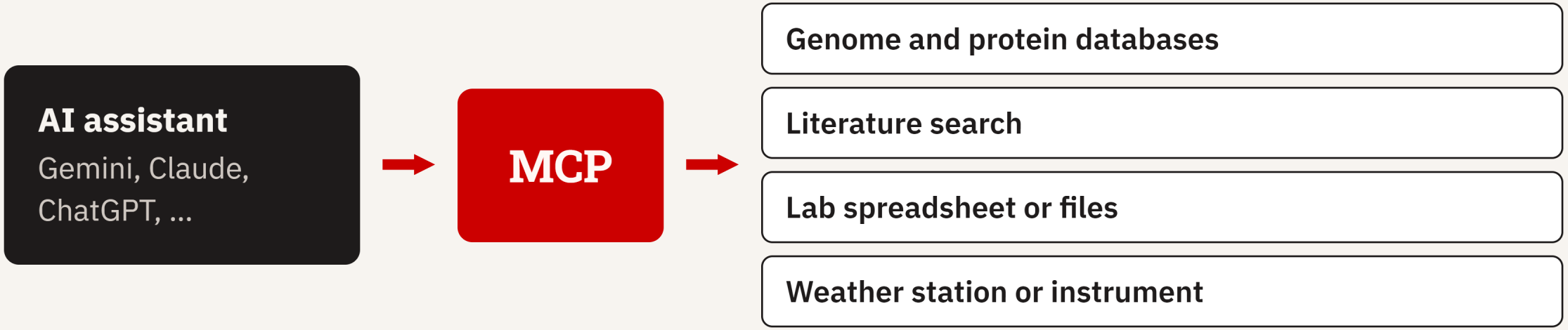
Databases
Your data, public data

Apps
Files, calendars, instruments

The model decides when to call a tool; the result goes back into its context.

MODEL CONTEXT PROTOCOL (MCP)

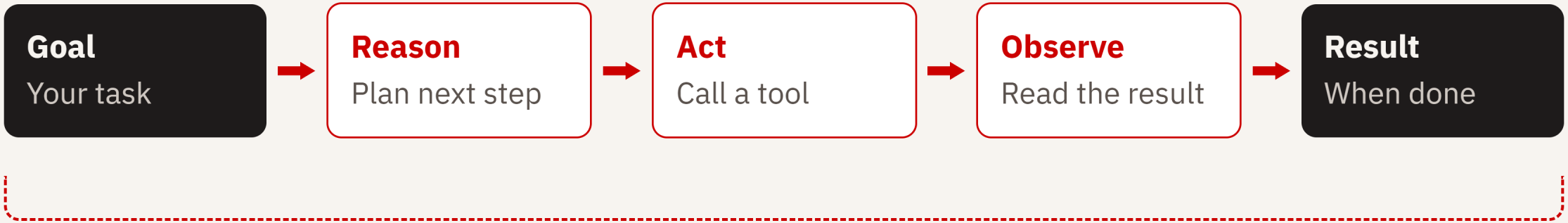
One standard plug for tools and data



An open standard: like USB-C, any assistant can connect to any tool that speaks it.

FROM ANSWERS TO ACTIONS

An agent is a model in a loop with tools



Persona + skills + tools through MCP + your documents, called again and again.

MULTI-AGENT SYSTEMS

A team of specialists, coordinated

Several agents, each with its own persona, tools and guardrails. **Orchestration** is how their work is divided and combined.

Orchestrator and workers
A coordinator splits the task and assigns the parts

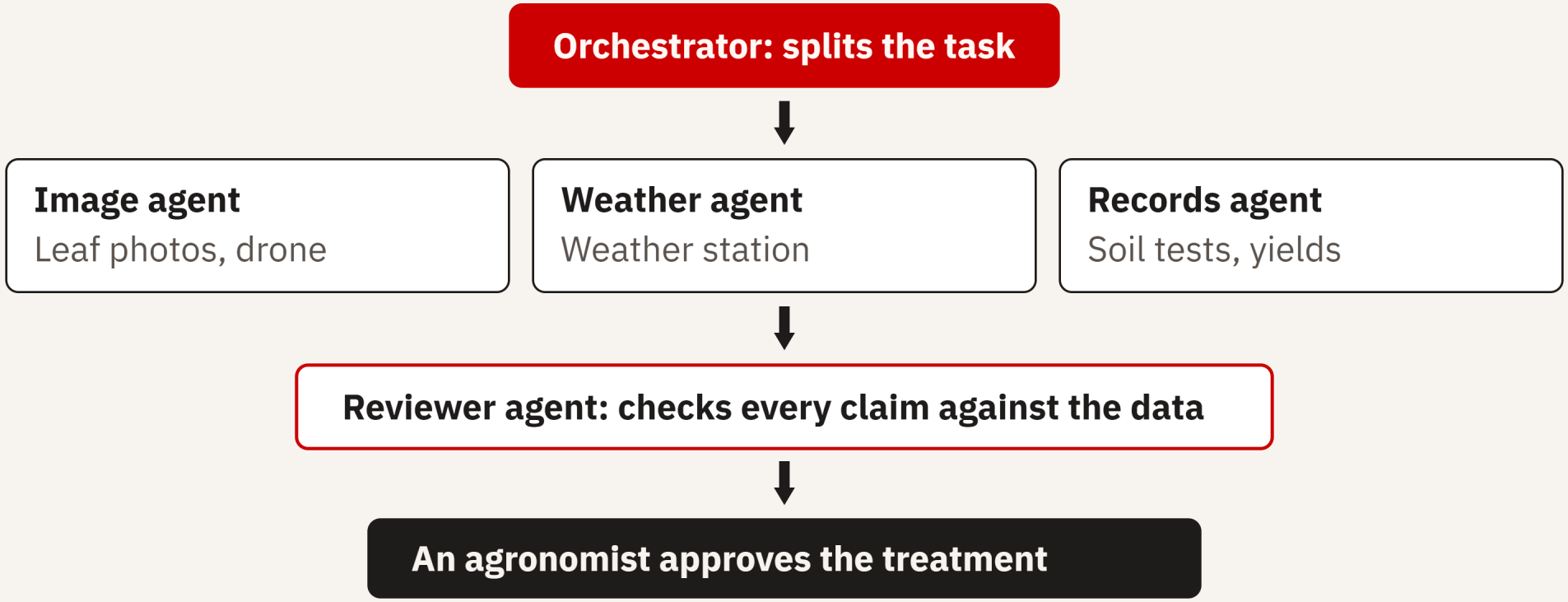
Sequential pipeline
Each agent’s output is the next one’s input

Parallel
Several agents work at once; results are merged

Critic loop
One produces, another critiques, and they repeat

EXAMPLE · AGRICULTURE

“Is this coffee plot at risk of leaf rust?”



MULTI-AGENT SYSTEMS

More agents, more to check

What you gain

- Focused specialists
- Work done in parallel
- Built-in review

What it costs

- Errors compound from agent to agent
- Harder to debug
- More calls, repeated context
- Handoffs must be clearly defined

Match the model to the job

- **Strong model:** plan and review
- **Small or local model:** narrow tasks
- **Plain code:** fixed steps

Decide where a person approves before the next step.

ACTIVITY B · 8 MIN

Your two notebooks from Activity A

You are the orchestrator

1 · Analysis Planner	“Plan an analysis that trains and evaluates a leaf-disease classifier on the beans dataset described in the README.”
2 · Critical Reviewer	“Review this analysis plan: [paste the Planner’s plan]”
3 · Analysis Planner	“A reviewer raised these issues: [paste the critique]. Revise your plan.”

Keep the Planner chat open for Part 5. Show of hands: at which step should a person decide?

PART 3

Grounding AI in your own documents

RAG, fine-tuning, and asking questions of your own papers

NC STATE UNIVERSITY

RAG or fine-tuning?

RETRIEVAL-AUGMENTED GENERATION

An open-book exam

- Finds relevant passages in your documents and puts them in the context
- Answers can cite their sources
- Update it by adding documents

e.g., NotebookLM (Activities A–C)

FINE-TUNING

Studying for the exam

- Retrains the model's weights on your data
- Teaches a new skill or style
- Needs data, compute and expertise

e.g., a vision model adapted to local crop diseases (Session 4)

Rule of thumb: need facts → RAG. Need a new skill → fine-tune.

ACTIVITY C · 12 MIN

notebooklm.google.com

Ask your own papers

1

Create a notebook and upload 2–3 papers from your area

2

Ask a question you would ask a colleague

3

Click each citation: does the source really say that?

4

Generate some other media content

PART 4

Working with AI

Modes of collaboration, three skills, and what it means for learning

NC STATE UNIVERSITY

THREE MODES OF WORKING WITH AI

Self-automator, centaur or cyborg?

Self-automator

DELEGATE EVERYTHING

The AI does the whole task; you accept the result

Centaur

DIVIDE THE WORK

You decide which parts the AI does and do the rest yourself

Cyborg

WORK TOGETHER CLOSELY

You and the AI iterate in short back-and-forth loops

Choose the mode on purpose, and notice when you slip into self-automator.

Three skills for working with AI

1

Specify

State the goal, data, units, constraints and what a reasonable answer looks like. Ask for a plan and its assumptions before any result.

2

Co-design

Ask for alternatives and their trade-offs. Have it argue against your choice and list every decision it made. You sign off.

3

Evaluate

Test on cases where you know the answer. Check units and magnitudes. Get a second critique. Open every source.

FOR EDUCATORS

From writing code to evaluating it

Before generative AI*



After generative AI*



Students must move from **expert writers** of code to **expert evaluators and good co-designers**.

* Estimates

FOR EDUCATORS · OFFLOADING VS. PRODUCTIVE STRUGGLE

AI can skip the thinking that builds skill, and we don't notice

Learning: 52 programmers learn a new Python library

Quiz score afterwards



Biggest gap on **debugging**. **How** they used AI mattered: asked it to explain → 65–86%; let it write the code → 24–39%.

Productivity: 16 experienced developers, 246 real tasks

Effect of AI on their speed



Even experts misjudge what AI does for them.

Design activities where students ask AI to explain, not to do, and protect the struggle that builds skill.

NC STATE UNIVERSITY

Shen & Tamkin, Anthropic (Jan 2026), "How AI assistance impacts the formation of coding skills" · METR (Jul 2025), randomized trial of early-2025 AI tools

FOR EDUCATORS

Assess the process, not just the product

Decompose
Grade the plan and the prompts

→

Generate
Ask students to explain and defend design decisions

→

Verify
Grade the checking: tests run, errors found, sources opened

Discussion: what does this look like in your own course or lab?

PART 5 · ACTIVITY D · 20 MIN

NotebookLM + colab.research.google.com

Plan, co-design, evaluate: bean leaf disease

THE PROMPT

“Turn your revised plan into steps I can paste into Gemini in Google Colab. The data is **beans128.npz**, uploaded to Colab and loaded as described in the README. The steps should: **1)** explore the data (images per class, sample photos); **2)** train a simple image classifier; **3)** evaluate it on the test split (accuracy, confusion matrix, examples it got wrong). List your assumptions.”

1 · Plan

Paste it into your Planner notebook chat from Activity B. Copy the steps.

2 · Run

New Colab notebook: upload beans128.npz, open Gemini, paste the steps.

3 · Evaluate

Use the next slide’s checklist. What did it decide alone?

Then type only “classify the images in beans128.npz” in a new notebook and compare.

Data: Makerere AI Lab “beans”, 128×128 copy · beans128.npz + README.md

Evaluation checklist

- Did it do what the plan said?
- What did it decide without telling you?
- Do the units and magnitudes make sense?
- Does one number survive a check by hand?
- Can you open and confirm every source?
- Would a colleague accept it as is?

What to take home

Context is everything
What the model sees decides what it says

You stay in charge
Specify, co-design and evaluate, especially with agents

Check before you trust
Confidence is not correctness

Next: Session 2 · Inside the Black Box: Foundations of Deep Learning

¡Gracias! Questions?

Edgar Lobaton · ejlobato@ncsu.edu
Applied AI Initiative, NC State College of Engineering

NC STATE UNIVERSITY